

Rong Wei

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EDUCATION

University of Illinois Urbana-Champaign

June 2025 – Present

Master of Science in Computer Science

- GPA: 3.83/4.00
- Coursework: Applied Machine Learning, Deep Learning for Healthcare, Statistical Modeling, Data Curation, Database Systems

Western Washington University

Sept 2020 – June 2024

Bachelor of Science in Computer Science

- GPA: 3.61/4.00
- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, Database Systems

EXPERIENCE

TradingFlow

July 2024 – July 2025

Full Stack Internship, Redmond, WA

- Engineered a full-stack options data platform using **Next.js**, **React**, **TypeScript**, enabling scalable visualization and interaction with large financial datasets.
- Designed and deployed high-performance backend APIs using **ClickHouse**, supporting efficient querying and user-driven data downloads.
- Integrated **Stripe** and authentication systems (Clerk) for secure subscription management and user access control.
- Optimized frontend rendering and backend query performance, improving responsiveness and contributing to a **30% increase in user engagement**.

PROJECTS

AI Weight Tracker App

July 2026 – Present

React Native, Expo, Firebase, TypeScript, OpenRouter AI

- Built a bilingual weight-loss and calorie tracking app using **React Native** and **Expo** for iOS and web platforms.
- Integrated **Firebase Authentication** and **Firestore** for secure accounts, persistent cloud syncing, and real-time updates.
- Developed AI-powered calorie estimation using the **OpenRouter API** with natural-language food input and automated meal analysis.

MedMod: Multimodal Medical Prediction Pipeline

Jan 2026 – Apr 2026

PyTorch, PyHealth, Python

- Reproduced a MedFuse-based multimodal model integrating EHR and imaging data for clinical prediction tasks.
- Built a robust data pipeline handling missing modalities using fallback strategies and dynamic feature selection.
- Conducted ablation studies to evaluate model components, achieving **AUROC $\approx$ 0.63** and analyzing performance trade-offs.

Academic World Dashboard

June 2025 – Aug 2025

Dash, MySQL, MongoDB, Neo4j

- Developed an interactive analytics platform to explore faculty, publications, and institutional research metrics.
- Built visualizations and comparison tools for keyword search, ranking, and side-by-side university analysis.
- Optimized multi-database queries using indexing, transactions, and efficient data integration across SQL and NoSQL systems.

Anniversary Report Tool | GeoEngineers | Corporate Alliance Program

Sept 2023 – June 2024

Node.js, React, PostgreSQL, AWS

- Engineered an automated employee milestone tracking system using React and PostgreSQL, streamlining HR workflows.
- Deployed the full-stack application on **AWS**, ensuring scalability, reliability, and secure organizational access.
- Developed a real-time notification system with Node.js to track anniversaries and milestones, improving engagement and visibility.

SKILLS

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, React Native, Next.js, Node.js, PyTorch, Dash, Expo, Jest

Databases: MySQL, PostgreSQL, MongoDB, Neo4j, ClickHouse, Firebase Firestore

Cloud & Developer Tools: AWS, Docker, Git, Linux, Stripe API, REST APIs

Concepts: Full-Stack Development, API Design, Data Engineering, Machine Learning, Mobile App Development